SIMATS ENGINEERING

ASSESSMENT TOOL 1: Simulation Task

COURSE NAME: **Embedded Systems**

COURSE CODE: **ECA1421**

COURSE OUTCOME COVERED

CO2: ***Apply Embedded C programming concepts, including structured design, real-time constraint handling, and SystemC-based modeling, to develop and analyze embedded applications for industrial use cases. (BL5)***

Name of the Student **: Siripireddy Hari chandana**

Register Number **: 192372136**

Create an Embedded C simulation for a vibration monitoring system using real-time processing

Requirements Analysis & Conceptual Understanding (20)	System Design & Simulation Model Development(25)	Implementation & Tool Usage (20)	Testing, Validation & Result Analysis (20)	Documentation(15)	Total(100)

Code of Conduct:

I Siripireddy Harichandana(19237216)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

AIM : To create an Embedded C simulation for a vibration monitoring system using the AT89C51 microcontroller, where the system detects vibration in real time and provides an alert through an LED and a buzzer.

DESIGN :

- The system is designed using the **AT89C51 (8051) microcontroller** as the main processing unit.
- A **Logic State** component in Proteus is used to simulate the vibration sensor and is connected to **Port P1.0**.
- An **LED** connected to **Port P2.0** provides a visual indication of vibration.
- A **buzzer** connected to **Port P2.1** provides an audible alert when vibration is detected.
- An **11.0592 MHz crystal oscillator** with **two 33 pF capacitors** provides the clock signal for the microcontroller.
- A **reset circuit** consisting of a **10 µF capacitor** and **10 kΩ resistor** ensures proper initialization during power-on.
- The design enables **real-time vibration detection** with both visual and audible alerts.

REQUIREMENTS :

Hardware Requirements

- AT89C51 (8051) Microcontroller
- 11.0592 MHz Crystal Oscillator
- Two 33 pF Capacitors
- One 10 µF Capacitor
- One 10 kΩ Resistor
- One 220 Ω Resistor
- One Red LED
- One 5 V Buzzer
- Logic State (used as a vibration sensor in Proteus)
- +5 V Power Supply

- Ground (GND)

Software Requirements

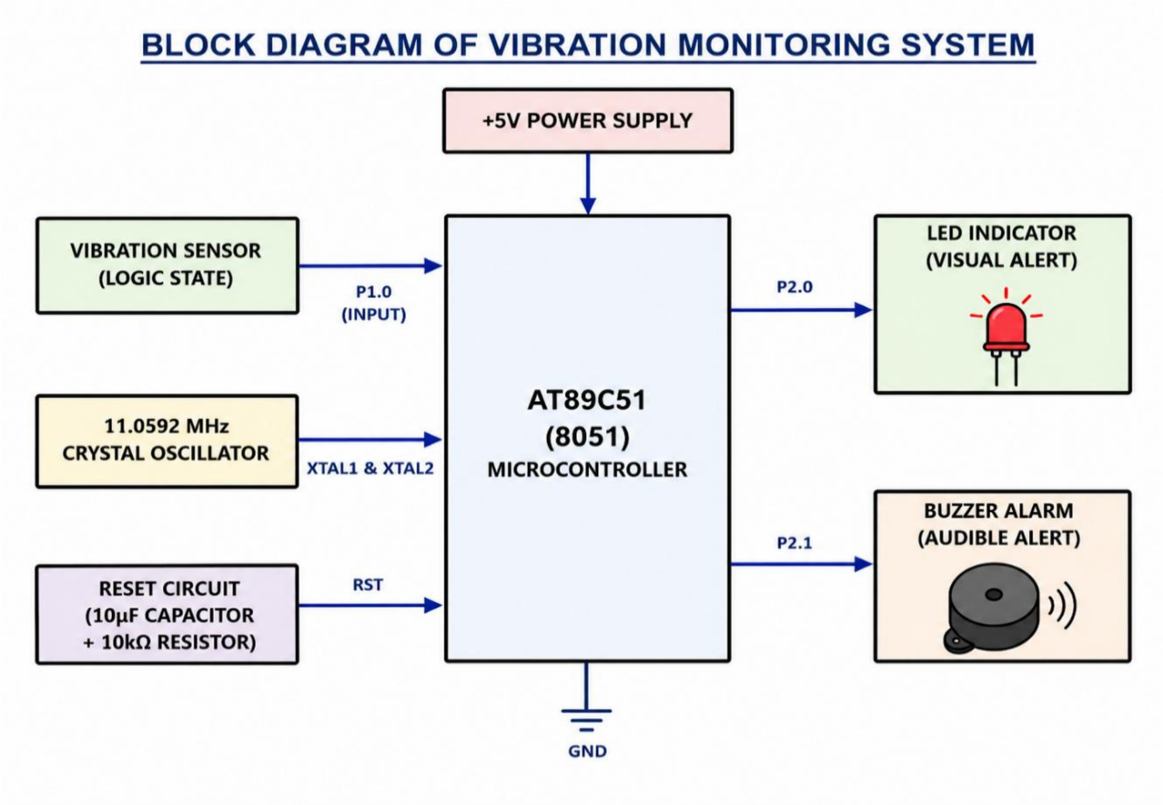
- Keil µVision IDE
- Proteus 8 Professional
- Embedded C Programming Language
- Windows Operating System (Windows 10/11 recommended)

Embedded C:

```
#include <reg51.h>

sbit VIBRATION = P1^0;

sbit LED       = P2^0;

sbit BUZZER    = P2^1;

void main(){

   VIBRATION = 1;      // Configure P1.0 as input

   LED = 0;            // Initially OFF

   BUZZER = 0;         // Initially OFF

   while(1)    {

      if(VIBRATION == 1)       {

         LED = 1;        // LED ON

         BUZZER = 1;     // Buzzer ON

      } else {

         LED = 0;        // LED OFF

         BUZZER = 0;     // Buzzer OFF

      }

   }
```

}

BLOCK DIAGRAM:



EXPLANATION:

1. +5V Power Supply

- Supplies 5V DC to the microcontroller and circuit.

2. Vibration Sensor (Logic State)

- Connected to P1.0.
- Logic 0: No vibration.
- Logic 1: Vibration detected.

3. 11.0592 MHz Crystal Oscillator

- Provides the clock signal for the AT89C51.

4. Reset Circuit

- Resets the microcontroller during power ON for proper startup.

5. AT89C51 Microcontroller

- Reads the sensor input and controls the LED and buzzer.

6. LED Indicator

- Connected to P2.0.
- Turns ON when vibration is detected.

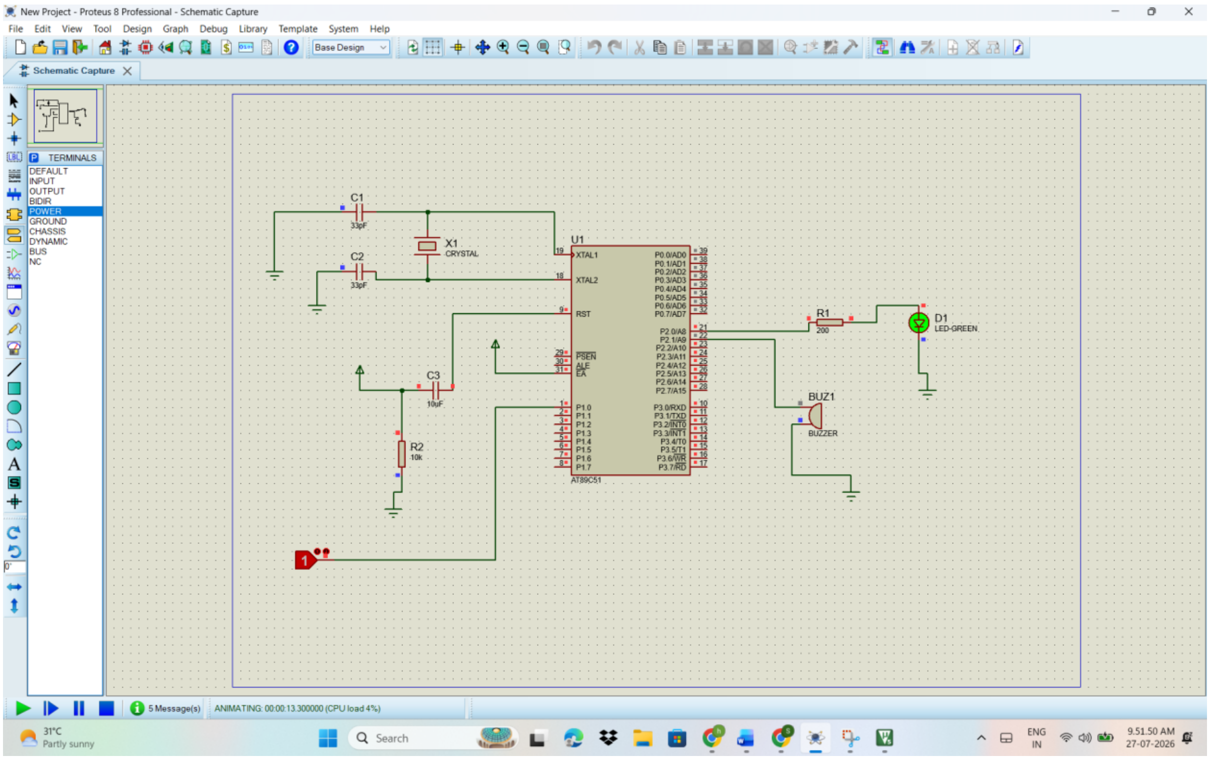
7. Buzzer

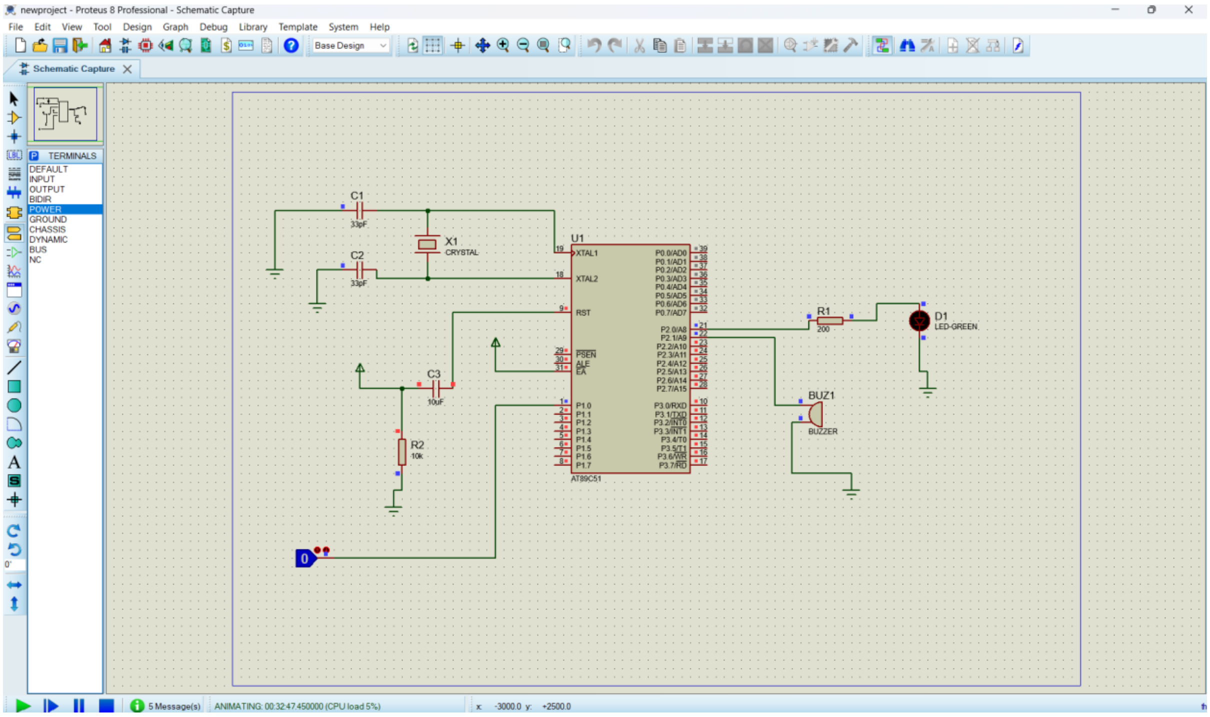
- Connected to P2.1.
- Sounds when vibration is detected.

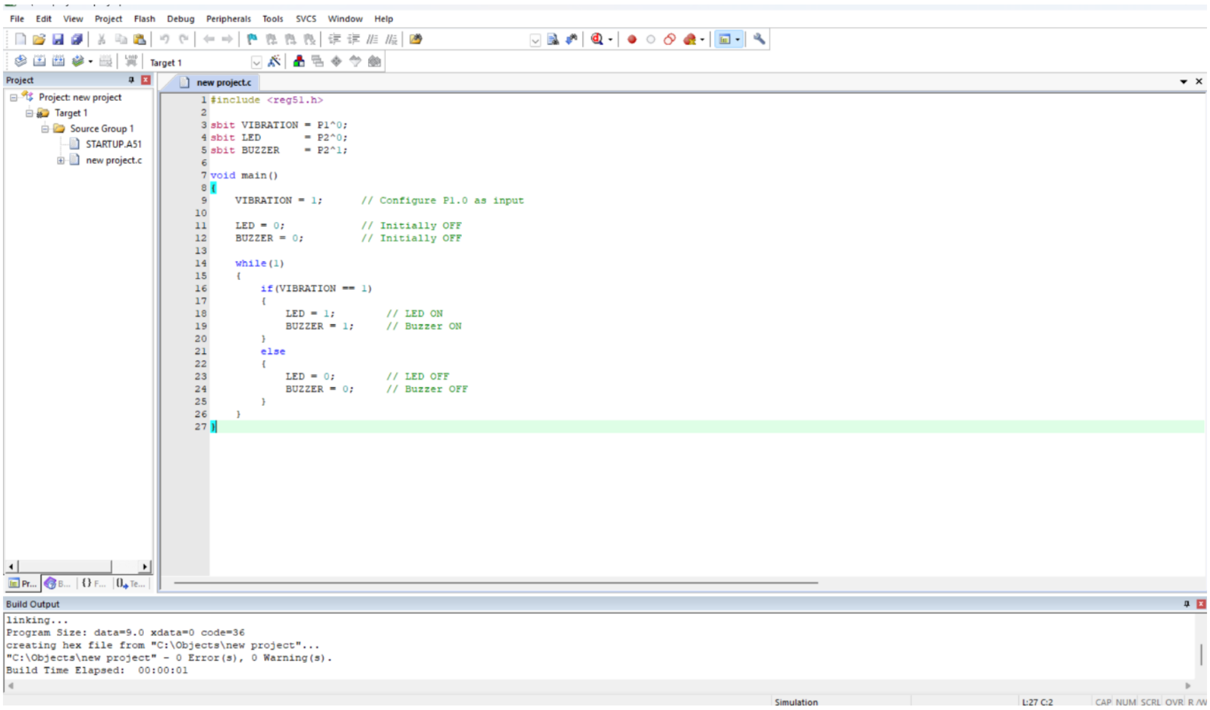
Working

1. The +5V supply powers the system.
2. The crystal oscillator and reset circuit initialize the AT89C51.
3. The microcontroller continuously checks the sensor at P1.0.
4. If the input is Logic 0, the LED and buzzer turn ON.
5. If the input is Logic 1, the LED and buzzer turn OFF.
6. The process repeats continuously for real-time vibration monitoring.

OUTPUT:




newproject - Proteus 8 Professional - Schematic Capture
File Edit View Tool Design Graph Debug Library Template System Help
Base Design
Schematic Capture
TERMINALS
DEFAULT
INPUT
OUTPUT
BIDIR
POWER
GROUND
CHASSIS
DYNAMIC
BUS
NC
C1
33pF
C2
33pF
X1
CRYSTAL
U1
XTAL1
XTAL2
RST
PSEN
ALE
EA
AT89C51
C3
10uF
R2
10k
R1
200
D1
LED-GREEN
BUZ1
BUZZER
5 Message(s)
ANIMATING: 00:32:47.450000 (CPU load 5%)


File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Target 1
Project
Project: new project
Target 1
Source Group 1
STARTUP.A51
new project.c
#include <reg51.h>
sbit VIBRATION = P1^0;
sbit LED = P2^0;
sbit BUZZER = P2^1;
void main()
{
VIBRATION = 1; // Configure P1.0 as input
LED = 0; // Initially OFF
BUZZER = 0; // Initially OFF
while(1)
{
if(VIBRATION == 1)
{
LED = 1; // LED ON
BUZZER = 1; // Buzzer ON
}
else
{
LED = 0; // LED OFF
BUZZER = 0; // Buzzer OFF
}
}
}
Build Output
linking...
Program Size: data=9.0 xdata=0 code=36
creating hex file from "C:\Objects\new project"...
"C:\Objects\new project" - 0 Error(s), 0 Warning(s).
Build Time Elapsed: 00:00:01
Simulation
L:27 C:2